

Bell-State Robustness to Routing-Induced Noise in Quantum Teleportation, with Error Mitigation (Linear ZNE)

Theory and Mathematical Derivations : Stages 1–4

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Abstract

This document consolidates the theoretical background and hand-derived mathematics behind the four completed stages of the project. Stage 1 establishes the density-matrix formalism for ideal teleportation. Stage 2 derives closed-form fidelity expressions for all four noise channels considered in the proposal (bit-flip, phase-flip, depolarizing, amplitude-damping) using the general noisy-teleportation superoperator. Stage 3 quantifies the extra SWAP overhead incurred by a hardware-forced qubit layout relative to a connectivity-respecting one, using the $[F \approx (1-p)^4]$ fidelity scaling. Stage 4 derives the Linear and Quadratic Zero-Noise Extrapolation (ZNE) mitigated fidelities from simulated data, and confirms that the Linear-ZNE result matches the Richardson formula stated in the project proposal. Each stage is presented with its full derivation so the mathematics can be checked independently of the accompanying simulation code.

1 Theoretical Background

Quantum teleportation transfers an unknown single-qubit state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (1)$$

from Alice to Bob using one of the four Bell states as a shared entanglement resource,

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle), \quad (2)$$

together with two bits of classical communication. In the noise-free case, the protocol reproduces $|\psi\rangle$ at Bob's side exactly, for any of the four Bell resources.

Noisy states are described with the density matrix $\rho = |\psi\rangle\langle\psi|$, and a general noise process acts via Kraus operators,

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger, \quad \sum_k E_k^\dagger E_k = I. \quad (3)$$

Using the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (4)$$

the four channels studied are:

$$\text{Bit-flip: } E_0 = \sqrt{1-p}I, \quad E_1 = \sqrt{p}X, \quad (5)$$

$$\text{Phase-flip: } E_0 = \sqrt{1-p}I, \quad E_1 = \sqrt{p}Z, \quad (6)$$

$$\text{Depolarizing: } E_0 = \sqrt{1-\frac{3p}{4}}I, \quad E_{1,2,3} = \sqrt{\frac{p}{4}}\{X, Y, Z\}, \quad (7)$$

$$\text{Amplitude-damping: } E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}. \quad (8)$$

Performance is quantified with fidelity $F = \langle \psi | \rho_{\text{out}} | \psi \rangle$, benchmarked against the classical (no-entanglement) limit

$$F_{\text{classical}} = \frac{2}{3}. \quad (9)$$

Routing on real chips forces SWAP-gate insertion when interacting qubits are not adjacent, adding real two-qubit gate error even though an ideal SWAP is noiseless. The worst-fidelity case is mitigated using Linear Zero-Noise Extrapolation (ZNE), built on unitary gate folding $U \rightarrow U(U^\dagger U)^k$, which leaves the ideal circuit unchanged while scaling up the effective hardware noise.

The four stages below work through this theory in full, hand-verified detail.

2 Stage 1: Ideal Teleportation in Density-Matrix Formalism

Before introducing noise, we establish the density-matrix formalism for the ideal teleportation protocol. This formalism is used consistently throughout Stages 1 and 2.

The input state is given by the density matrix:

$$\rho_\psi = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}. \quad (10)$$

For a Bell state, say $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$, the entangled resource density matrix is:

$$\rho_{\text{ent}} = |\Psi^+\rangle\langle\Psi^+|. \quad (11)$$

The total initial state of the three-qubit system (input qubit q_0 , Alice's qubit q_1 , Bob's qubit q_2) is:

$$\rho_{\text{in}} = \rho_\psi \otimes \rho_{\text{ent}}. \quad (12)$$

The teleportation circuit applies a unitary evolution U_{tel} (the sequence of CNOTs, Hadamard, and measurements, with classical feed-forward corrections) followed by a partial trace over Alice's qubits (q_0 and q_1):

$$\rho_{\text{out}} = \text{Tr}_{1,2} \left[U_{\text{tel}} (\rho_\psi \otimes \rho_{\text{ent}}) U_{\text{tel}}^\dagger \right]. \quad (13)$$

For the standard teleportation protocol, Eq. (13) simplifies to:

$$\boxed{\rho_{\text{out}} = \rho_\psi}. \quad (14)$$

Thus, the fidelity of the ideal protocol is:

$$F = \langle \psi | \rho_{\text{out}} | \psi \rangle = \langle \psi | \rho_\psi | \psi \rangle = 1. \quad (15)$$

This confirms that the teleportation circuit is correctly implemented and that, in the noiseless case, Bob receives the input state with perfect fidelity for all four Bell states. This density-matrix formalism provides the foundation for the noisy-channel analysis in Stage 2.

3 Stage 2: Closed-Form Fidelities for the Four Noise Channels

3.1 General Noisy-Teleportation Superoperator

Following the approach of Oh et al. (2002, Eq. 8), when noise is applied to the input qubit before teleportation, the output state is given by:

$$\mathcal{E}(\rho_{\text{out}}) = \text{Tr}_{1,2} \left[U_{\text{tel}} (\mathcal{E}(\rho_{\text{in}}) \otimes \rho_{\text{ent}}) U_{\text{tel}}^\dagger \right], \quad (16)$$

where $\mathcal{E}$ is the noise superoperator acting on the input qubit, and ρ_{ent} is the Bell-state resource.

For a Pauli channel of the form $\mathcal{E}(\rho) = (1-p)\rho + pA\rho A$ (where $A \in \{X, Y, Z\}$), Eq. (16) reduces to:

$$\mathcal{E}(\rho_{\text{out}}) = (1-p)\rho_{\text{in}} + pA\rho_{\text{in}}A. \quad (17)$$

This follows from the linearity of the partial trace and unitary evolution, and the fact that ideal teleportation is the identity map on the input state. The full reduction is provided in Appendix A.

3.2 Input state and general lemma

Take the input qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha = \frac{\sqrt{3}}{2}, \quad \beta = \frac{1}{2}, \quad (18)$$

normalized since $\alpha^2 + \beta^2 = \frac{3}{4} + \frac{1}{4} = 1$. Its density matrix is

$$\rho_\psi = |\psi\rangle\langle\psi| = \begin{pmatrix} \frac{3}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{1}{4} \end{pmatrix}, \quad \text{Tr}(\rho_\psi) = 1. \quad (19)$$

By the reduction in Eq. (17), teleportation fidelity under input-qubit noise reduces to a single-qubit calculation,

$$F(p) = \langle\psi| \mathcal{E}(\rho_\psi) |\psi\rangle. \quad (20)$$

For any single-Pauli-error channel $\mathcal{E}(\rho) = (1-p)\rho + pA\rho A$ with Hermitian unitary A , writing $A|\psi\rangle = |\phi\rangle$ gives $\langle\psi|A\rho_\psi A|\psi\rangle = |\langle\psi|\phi\rangle|^2 = \langle A \rangle^2$, so

$$\boxed{F(p) = (1-p) + p\langle A \rangle^2}. \quad (21)$$

For this input state,

$$\langle X \rangle = 2\alpha\beta = \frac{\sqrt{3}}{2}, \quad \langle Z \rangle = \alpha^2 - \beta^2 = \frac{1}{2}, \quad \langle Y \rangle = 0, \quad (22)$$

$$\langle X \rangle^2 = \frac{3}{4}, \quad \langle Z \rangle^2 = \frac{1}{4}, \quad \langle Y \rangle^2 = 0. \quad (23)$$

3.3 Bit-flip noise

With $E_0 = \sqrt{1-p}I$, $E_1 = \sqrt{p}X$:

$$X\rho_\psi X = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad \rho_{\text{BF}} = (1-p)\rho_\psi + pX\rho_\psi X = \begin{pmatrix} \frac{3}{4} - \frac{p}{2} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{1}{4} + \frac{p}{2} \end{pmatrix}. \quad (24)$$

Via the lemma with $\langle X \rangle^2 = 3/4$: $F_{\text{BF}}(p) = (1-p) + \frac{3}{4}p$. Direct evaluation of $\langle \psi | \rho_{\text{BF}} | \psi \rangle$ gives the same result:

$$\boxed{F_{\text{BF}}(p) = 1 - \frac{p}{4}.} \quad (25)$$

3.4 Phase-flip noise

With $E_0 = \sqrt{1-p}I$, $E_1 = \sqrt{p}Z$:

$$Z\rho_\psi Z = \begin{pmatrix} 3/4 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & 1/4 \end{pmatrix}, \quad \rho_{\text{PF}} = \begin{pmatrix} \frac{3}{4} & \frac{\sqrt{3}}{4}(1-2p) \\ \frac{\sqrt{3}}{4}(1-2p) & \frac{1}{4} \end{pmatrix}. \quad (26)$$

The populations are unaffected; all p -dependence sits in the coherence term. Via the lemma with $\langle Z \rangle^2 = 1/4$, confirmed by direct evaluation:

$$\boxed{F_{\text{PF}}(p) = 1 - \frac{3p}{4}.} \quad (27)$$

3.5 Depolarizing noise

Using the proposal's convention $E_0 = \sqrt{1 - \frac{3p}{4}}I$, $E_{1,2,3} = \sqrt{\frac{p}{4}}\{X, Y, Z\}$, so that $p = 1$ corresponds to full depolarization to $I/2$. Computing $X\rho_\psi X$, $Y\rho_\psi Y$, $Z\rho_\psi Z$ and summing,

$$X\rho_\psi X + Y\rho_\psi Y + Z\rho_\psi Z = \begin{pmatrix} 5/4 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & 7/4 \end{pmatrix}, \quad (28)$$

$$\rho_{\text{DP}} = \left(1 - \frac{3p}{4}\right)\rho_\psi + \frac{p}{4} \begin{pmatrix} 5/4 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & 7/4 \end{pmatrix} = \begin{pmatrix} \frac{3}{4} - \frac{p}{4} & \frac{\sqrt{3}}{4}(1-p) \\ \frac{\sqrt{3}}{4}(1-p) & \frac{1}{4} + \frac{p}{4} \end{pmatrix}. \quad (29)$$

Via the lemma summed over the three Pauli errors, using Eq. (23), and confirmed directly:

$$\boxed{F_{\text{DP}}(p) = 1 - \frac{p}{2}.} \quad (30)$$

3.6 Amplitude-damping noise

With $E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}$, $E_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$:

$$\rho_{\text{AD}} = E_0 \rho_\psi E_0^\dagger + E_1 \rho_\psi E_1^\dagger = \begin{pmatrix} \frac{3}{4} + \frac{\gamma}{4} & \frac{\sqrt{3}}{4} \sqrt{1-\gamma} \\ \frac{\sqrt{3}}{4} \sqrt{1-\gamma} & \frac{1}{4}(1-\gamma) \end{pmatrix}. \quad (31)$$

Evaluating $F_{\text{AD}}(\gamma) = \langle \psi | \rho_{\text{AD}} | \psi \rangle$ term by term gives

$$F_{\text{AD}}(\gamma) = \frac{5}{8} + \frac{\gamma}{8} + \frac{3}{8} \sqrt{1-\gamma}. \quad (32)$$

As a sanity check, at $\gamma = 1$ (full decay to $|0\rangle$) this must equal $|\langle \psi | 0 \rangle|^2 = \alpha^2 = 3/4$; the formula gives $\frac{5}{8} + \frac{1}{8} + 0 = \frac{3}{4}$, confirming it.

3.7 Summary

Table 1: Closed-form fidelities for input state $\alpha = \sqrt{3}/2$, $\beta = 1/2$.

Channel	Fidelity $F(p)$
Bit-flip	$1 - p/4$
Phase-flip	$1 - 3p/4$
Depolarizing	$1 - p/2$
Amplitude-damping	$\frac{5}{8} + \frac{p}{8} + \frac{3}{8} \sqrt{1-p}$

Observations. Phase-flip is the most fragile channel for this input state (crossing $F_{\text{classical}} = 2/3$ at $p = 4/9$), consistent with $\langle Z \rangle^2$ being small; bit-flip and amplitude-damping never drop below the classical benchmark on $p, \gamma \in [0, 1]$, since this particular state is close to the $|0\rangle$ fixed point of amplitude damping and has large $\langle X \rangle^2$. This is a property of the chosen state, not of the channels in general, and is flagged explicitly for the final report.

4 Stage 3: Natural vs. Forced-Layout Teleportation Circuits

Convention. A three-qubit ket $|q_0 q_1 q_2\rangle$ lists qubits in order (q_0, q_1, q_2) : q_0 carries the input state, q_1 is Alice's half of the Bell pair, q_2 is Bob's half. Input state:

$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle, \quad q_1 = |0\rangle, \quad q_2 = |0\rangle.$$

4.1 Natural layout

On the physical chain $Q_0 - Q_1 - Q_2$ with $q_0 \rightarrow Q_0$, $q_1 \rightarrow Q_1$, $q_2 \rightarrow Q_2$, both required CNOTs ($q_1 \leftrightarrow q_2$ and $q_0 \leftrightarrow q_1$) land on adjacent qubits: **no SWAP is needed**.

The natural layout circuit uses exactly **2 two-qubit gates** (both CNOTs).

4.2 Forced layout

With $q_0 \rightarrow Q_0$, $q_2 \rightarrow Q_1$, $q_1 \rightarrow Q_2$ (Alice’s and Bob’s qubits swapped relative to natural), the second CNOT ($q_0 \leftrightarrow q_1$) now acts on the two *end* qubits: a **SWAP is required**.

4.2.1 SWAP-as-three-CNOTs identity

Using $\text{CNOT}|a, b\rangle = |a, a \oplus b\rangle$, the identity $\text{SWAP}(a, b) = \text{CNOT}(a, b) \cdot \text{CNOT}(b, a) \cdot \text{CNOT}(a, b)$ is verified:

$$|0, 1\rangle \rightarrow |0, 1\rangle \rightarrow |1, 1\rangle \rightarrow |1, 0\rangle,$$

confirming the swap: **every SWAP costs three real two-qubit gates**, not one.

The forced layout therefore uses:

- 2 required CNOT gates, and
- 1 SWAP gate (implemented as 3 CNOTs),

for a total of **5 two-qubit gates**.

4.3 Fidelity Scaling with Gate Count

Following Rindell et al. (2023, Section 2.2), for a circuit with l two-qubit gates each with error probability p , the fidelity scales as:

$$F \approx (1 - p)^l. \quad (33)$$

Applying this to our two layouts:

$$F_{\text{natural}} = (1 - p)^2, \quad F_{\text{forced}} = (1 - p)^5. \quad (34)$$

The relative fidelity drop due to routing overhead is therefore:

$$\frac{F_{\text{natural}} - F_{\text{forced}}}{F_{\text{natural}}} = 1 - (1 - p)^3 \approx 3p. \quad (35)$$

For typical IBM gate errors ($p \approx 0.01$), this gives approximately 3% fidelity loss, matching the proposal’s original prediction. With the actual FakeSherbrooke calibration errors ($p \approx 0.0075$), the predicted drop is approximately 2.25%.

4.4 Ideal-Case Check

In the noiseless case ($p = 0$), both layouts give:

$$F_{\text{natural}} = F_{\text{forced}} = 1, \quad (36)$$

confirming that the SWAP is a mathematical no-op on the ideal state and only introduces error through real hardware noise. The routing overhead is purely a hardware-noise effect, quantified in simulation.

4.5 Circuit Diagrams

The circuits for both layouts, as transpiled onto the FakeSherbrooke coupling map, are shown below. The natural layout uses 2 two-qubit gates; the forced layout uses 5 (the extra 3 come from the SWAP).

[Insert circuit diagrams from code here]

5 Stage 4: Error Mitigation — Linear and Quadratic ZNE

5.1 Identity folding

In an ideal system, any unitary U satisfies $U^\dagger U = I$. Identity folding replaces U with

$$U_{\text{folded}} = U(U^\dagger U)^k, \quad k \in \{0, 1, 2, \dots\}, \quad (37)$$

which expands to $U_{\text{folded}} = U(I)^k = U$: mathematically, folding does not alter the ideal quantum state. On real hardware or noisy simulation, however, each additional physical gate introduces decoherence, so folding scales up the *effective* noise while leaving the ideal operation logically unchanged. Since the hardware error parameters are not known analytically, the fidelity is measured empirically at several folding scales and extrapolated to the zero-noise limit ($\lambda \rightarrow 0$).

5.2 Simulated data

$$\begin{array}{ll} \lambda_1 = 1, & F(\lambda_1) = 0.9476 \quad (k = 0) \\ \lambda_2 = 3, & F(\lambda_2) = 0.9027 \quad (\text{fold } 3\times, k = 1) \\ \lambda_3 = 5, & F(\lambda_3) = 0.8631 \quad (\text{fold } 5\times, k = 2) \end{array}$$

5.3 Linear ZNE

Modeling fidelity as $F(\lambda) = m\lambda + c$ and using $(\lambda_1, F(\lambda_1))$, $(\lambda_2, F(\lambda_2))$:

$$m = \frac{F(\lambda_2) - F(\lambda_1)}{\lambda_2 - \lambda_1} = \frac{0.9027 - 0.9476}{3 - 1} = \frac{-0.0449}{2} = -0.02245. \quad (38)$$

The negative slope indicates each $1 \times$ increase in folding decreases fidelity by 0.02245 (2.245%). Substituting back for the intercept,

$$0.9476 = (-0.02245)(1) + c \Rightarrow c = 0.97005. \quad (39)$$

Evaluating at zero noise, $F(\lambda) = -0.02245\lambda + 0.97005$:

$$\boxed{F_{\text{after mitigation}} = F(0) = 0.97005 \approx 0.9700.} \quad (40)$$

Cross-check against the proposal. The proposal's Richardson two-point formula (its Eq. 12) is

$$F_{\text{ZNE}} = \frac{3F(1) - F(3)}{2} = \frac{3(0.9476) - 0.9027}{2} = \frac{2.8428 - 0.9027}{2} = 0.97005, \quad (41)$$

which matches the slope-intercept result exactly, confirming the two formulations of Linear ZNE are algebraically equivalent and that both give the same mitigated fidelity.

5.4 Quadratic ZNE (Lagrange interpolation) — verification

To check whether non-linear noise effects matter, a 2nd-degree Lagrange polynomial through all three points is used:

$$P(\lambda) = F(\lambda_1)L_1(\lambda) + F(\lambda_2)L_2(\lambda) + F(\lambda_3)L_3(\lambda), \quad L_i(\lambda) = \prod_{j \neq i} \frac{\lambda - \lambda_j}{\lambda_i - \lambda_j}. \quad (42)$$

At $\lambda = 0$:

$$L_1(0) = \frac{(0-3)(0-5)}{(1-3)(1-5)} = 1.875, \quad L_2(0) = \frac{(0-1)(0-5)}{(3-1)(3-5)} = -1.25, \quad L_3(0) = \frac{(0-1)(0-3)}{(5-1)(5-3)} = 0.375. \quad (43)$$

As a consistency check, $L_1(0) + L_2(0) + L_3(0) = 1.875 - 1.25 + 0.375 = 1.0$ exactly. Weighting each fidelity:

$$\begin{aligned} \text{Term 1} &= 0.9476 \times 1.875 = 1.77675, \\ \text{Term 2} &= 0.9027 \times (-1.25) = -1.128375, \\ \text{Term 3} &= 0.8631 \times 0.375 = 0.3236625, \end{aligned}$$

$$P(0) = 1.77675 - 1.128375 + 0.3236625 = 0.9720375 \approx \boxed{0.9719}. \quad (44)$$

5.5 Conclusion

The quadratic extrapolation ($F = 0.9719$) deviates from the linear result ($F = 0.9700$) by less than 0.2%, verifying that higher-order non-linear noise effects are small over this folding range and that the Linear ZNE result reported for the worst-performing Bell-state/channel/layout combination is a reliable mitigated-fidelity estimate for the final report.

6 Overall Consistency Summary

- **Stage 1** establishes the density-matrix formalism for ideal teleportation, showing that $\rho_{\text{out}} = \rho_{\psi}$ with fidelity $F = 1$ for all four Bell states.
- **Stage 2** uses the general noisy-teleportation superoperator $\mathcal{E}(\rho_{\text{out}}) = \text{Tr}_{1,2}[U_{\text{tel}}(\mathcal{E}(\rho_{\text{in}}) \otimes \rho_{\text{ent}})U_{\text{tel}}^\dagger]$ and derives closed-form fidelities for all four noise channels, cross-checked against the Qiskit/QuTiP simulation sweep over $p \in [0, 0.75]$.
- **Stage 3** verifies the SWAP-as-three-CNOTs identity and quantifies routing overhead via the fidelity scaling $F \approx (1-p)^l$, giving $F_{\text{natural}} = (1-p)^2$ and $F_{\text{forced}} = (1-p)^5$, with a relative drop of approximately $3p$.
- **Stage 4** reproduces the proposal's own Richardson-extrapolation formula exactly from first principles, and additionally verifies the linear approximation against a quadratic (Lagrange) fit, confirming the chosen mitigation method is appropriate for the observed folding-noise data.

All four stages are mathematically self-consistent, correctly grounded in the equations and conventions fixed by the project proposal, and ready to be quoted directly in the final report.

A Appendix A: Reduction of the General Noise Superoperator to the Pauli-Channel Form

Starting from the general noisy-teleportation superoperator (Oh et al., 2002, Eq. 8):

$$\mathcal{E}(\rho_{\text{out}}) = \text{Tr}_{1,2} \left[U_{\text{tel}} (\mathcal{E}(\rho_{\text{in}}) \otimes \rho_{\text{ent}}) U_{\text{tel}}^\dagger \right]. \quad (45)$$

For a Pauli channel $\mathcal{E}(\rho) = (1 - p)\rho + pA\rho A$:

$$\mathcal{E}(\rho_{\text{out}}) = \text{Tr}_{1,2} \left[U_{\text{tel}} ((1 - p)\rho_{\text{in}} + pA\rho_{\text{in}}A) \otimes \rho_{\text{ent}} U_{\text{tel}}^\dagger \right] \quad (46)$$

$$= (1 - p) \text{Tr}_{1,2} \left[U_{\text{tel}} (\rho_{\text{in}} \otimes \rho_{\text{ent}}) U_{\text{tel}}^\dagger \right] + p \text{Tr}_{1,2} \left[U_{\text{tel}} (A\rho_{\text{in}}A \otimes \rho_{\text{ent}}) U_{\text{tel}}^\dagger \right]. \quad (47)$$

Since ideal teleportation transmits any input state perfectly:

$$\text{Tr}_{1,2} \left[U_{\text{tel}} (\rho_{\text{in}} \otimes \rho_{\text{ent}}) U_{\text{tel}}^\dagger \right] = \rho_{\text{in}}, \quad (48)$$

$$\text{Tr}_{1,2} \left[U_{\text{tel}} (A\rho_{\text{in}}A \otimes \rho_{\text{ent}}) U_{\text{tel}}^\dagger \right] = A\rho_{\text{in}}A. \quad (49)$$

Therefore:

$$\boxed{\mathcal{E}(\rho_{\text{out}}) = (1 - p)\rho_{\text{in}} + pA\rho_{\text{in}}A.} \quad (50)$$

This reduction holds for any Pauli channel ($A \in \{X, Y, Z\}$). For the depolarizing channel, the sum over all three Pauli operators gives the corresponding expression. For amplitude-damping, which is not a Pauli channel, this reduction does not apply and the full Kraus-operator calculation is used.